

Bookstore Data Warehouse

End-to-End Project Documentation

OLTP → Star Schema → SSIS ETL → SSAS Cube

Tool	SQL Server + Visual Studio (SSIS / SSAS)
Source DB	gravity_books (OLTP)
Target DWH	Bookstore_DWH
Grain	One row per order line (order_line)
Date	April 2026

1. Project Overview

This project builds a full Data Warehouse for a bookstore system. The source is an OLTP PostgreSQL/SQL Server database (gravity_books) containing transactional data about books, authors, customers, and orders. The goal is to transform this data into a Star Schema optimised for analytical queries.

Phase	Tool	Output
Source OLTP	gravity_books DB	11 normalized tables
Modeling	Star Schema Design	1 Fact + 6 Dims + 1 Bridge
DWH Creation	SQL Server	Bookstore_DWH database
ETL	SSIS	8 Packages — data loaded
Analysis	SSAS	Bookstore_DWH_View.cube

2. OLTP Source Schema (gravity_books)

The source database contains 11 tables organised around books, customers, and orders:

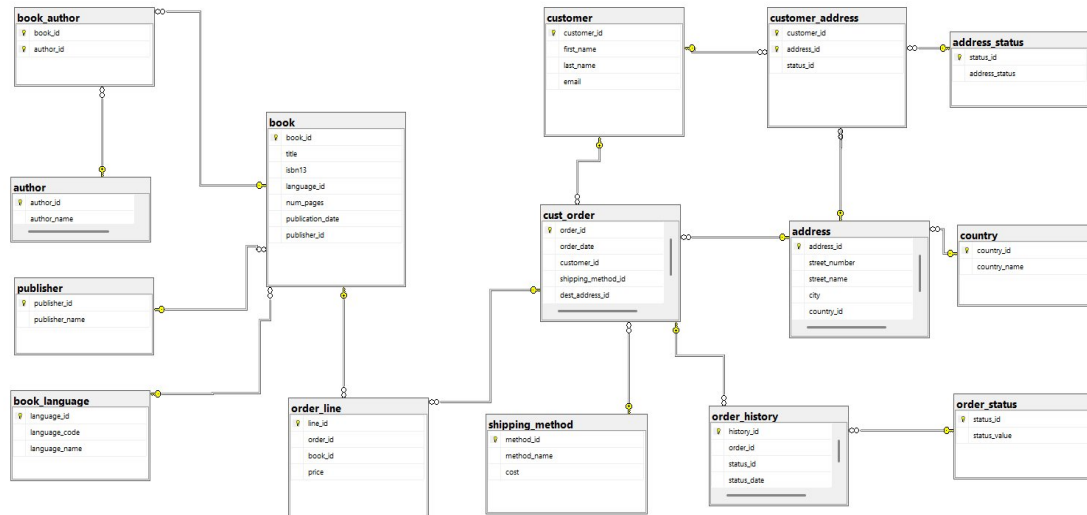


Figure 1 — OLTP Entity Relationship Diagram (gravity_books)

Table	Description
book	Core book data — title, ISBN, pages, publication date
author	Author names
book_author	Many-to-many junction between books and authors
book_language	Language lookup
publisher	Publisher names
customer	Customer — name and email
address	Street, city, country
country	Country lookup
cust_order	Order header — date, customer, shipping, address
order_line	Order lines — book and price
order_history	Status history per order
order_status	Status values (pending, shipped, delivered...)
shipping_method	Shipping method name and cost

3. Star Schema Design

Grain: One row per order line — the most granular level of a sale. Every row in fact_sales represents one book sold in one order.

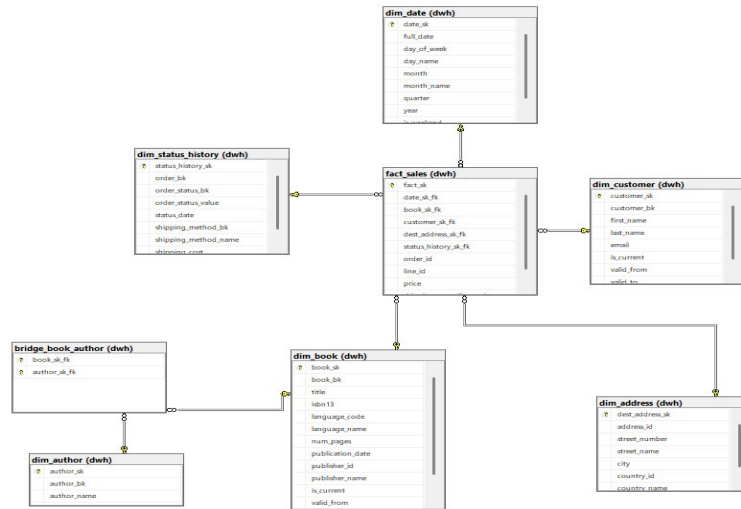


Figure 2 — DWH Star Schema Diagram (Bookstore_DWH)

Table	Type	Description
fact_sales	FACT	One row per order line — price, shipping cost
dim_date	DIM	Generated calendar — year, quarter, month, week, day
dim_book	DIM SCD2	Book details — title, ISBN, language, publisher (tracks changes)
dim_customer	DIM SCD2	Customer — name, email (tracks changes)
dim_address	DIM	Delivery address — street, city, country
dim_status_history	DIM	Latest order status + shipping method and cost
bridge_book_author	BRIDGE	Resolves many-to-many between books and authors
dim_author	DIM	Author name

SCD Type 2 Strategy

dim_book and dim_customer use Slowly Changing Dimension Type 2. When an attribute changes (e.g. customer email), the old record is expired (is_current = 0, valid_to = today) and a new record is inserted (is_current = 1, valid_from = today, valid_to = 9999-12-31). This preserves the full history of changes.

4. DWH Table Creation — SQL Scripts

All tables are created in the **dwh** schema inside the **Bookstore_DWH** database. The creation order respects foreign key dependencies: dimensions first, fact last.

dim_date

```
CREATE TABLE dwh.dim_date (  
    date_sk      INT      NOT NULL PRIMARY KEY,  
    full_date    DATE      NOT NULL,  
    day_of_week  TINYINT  NOT NULL,  
    day_name     NVARCHAR(10),  
    month        TINYINT  NOT NULL,  
    month_name   NVARCHAR(10),  
    quarter      TINYINT  NOT NULL,  
    year         SMALLINT NOT NULL,  
    is_weekend   BIT       NOT NULL DEFAULT 0  
);
```

dim_book (SCD2)

```
CREATE TABLE dwh.dim_book (  
    book_sk      INT IDENTITY(1,1) PRIMARY KEY,  
    book_bk      INT          NOT NULL,    -- NK  
    title        NVARCHAR(1000) NOT NULL,  
    isbn13       NVARCHAR(20),  
    language_code NVARCHAR(10),  
    language_name NVARCHAR(100),  
    num_pages     INT,  
    publication_date DATE,  
    publisher_id  INT,  
    publisher_name NVARCHAR(1000),  
    is_current    BIT NOT NULL DEFAULT 1,    -- SCD2  
    valid_from    DATE NOT NULL DEFAULT CAST(GETDATE() AS DATE),  
    valid_to      DATE NOT NULL DEFAULT '9999-12-31'  
);
```

dim_customer (SCD2)

```
CREATE TABLE dwh.dim_customer (  
    customer_sk INT IDENTITY(1,1) PRIMARY KEY,  
    customer_bk INT          NOT NULL,    -- NK  
    first_name  NVARCHAR(100) NOT NULL,  
    last_name   NVARCHAR(100) NOT NULL,  
    email       NVARCHAR(200),  
    is_current  BIT NOT NULL DEFAULT 1,  
    valid_from  DATE NOT NULL DEFAULT CAST(GETDATE() AS DATE),  
    valid_to    DATE NOT NULL DEFAULT '9999-12-31'  
);
```

fact_sales

```
CREATE TABLE dwh.fact_sales (  
    fact_sk      INT IDENTITY(1,1) PRIMARY KEY,  
    date_sk_fk    INT          NOT NULL,  
    book_sk_fk    INT          NOT NULL,
```

```
customer_sk_fk          INT          NOT NULL,
dest_address_sk_fk      INT          NOT NULL,
status_history_sk_fk    INT          NOT NULL,
order_id                INT          NOT NULL,  -- DD
line_id                 INT          NOT NULL,  -- DD
price                   DECIMAL(10,2) NOT NULL,
shipping_cost_allocated DECIMAL(10,2),
FOREIGN KEY (date_sk_fk) REFERENCES dwh.dim_date(date_sk),
FOREIGN KEY (book_sk_fk) REFERENCES dwh.dim_book(book_sk),
FOREIGN KEY (customer_sk_fk) REFERENCES dwh.dim_customer(customer_sk),
FOREIGN KEY (dest_address_sk_fk) REFERENCES dwh.dim_address(dest_address_sk),
FOREIGN KEY (status_history_sk_fk) REFERENCES dwh.dim_status_history(status_history_sk)
);
```

5. SSIS ETL Packages

Eight SSIS packages load data from the OLTP source (gravity_books) into the DWH. Each package has two Connection Managers: **gravity_books** (source) and **LocalHost.Bookstore_DWH** (destination).

#	Step	Details
1	dim_date.dtsx	Execute SQL Task only — generates calendar rows 2015–2035
2	dim_author.dtsx	OLE DB Source → OLE DB Destination (9,235 rows)
3	dim_book.dtsx	Source → Lookup → Conditional Split → SCD2 logic (11,127 rows)
4	dim_customer.dtsx	Source → Lookup → Conditional Split → SCD2 logic (2,000 rows)
5	dim_address.dtsx	Source (JOIN address+country) → OLE DB Destination
6	dim_status_history.dtsx	Source (latest status per order) → OLE DB Destination (7,544 rows)
7	dim_bridge_b.dtsx	Source → Lookup dim_book → Lookup dim_author → Destination (17,642 rows)
8	fact_sales.dtsx	Source → 5 Lookups → Derived Column → Destination (15,385 rows)

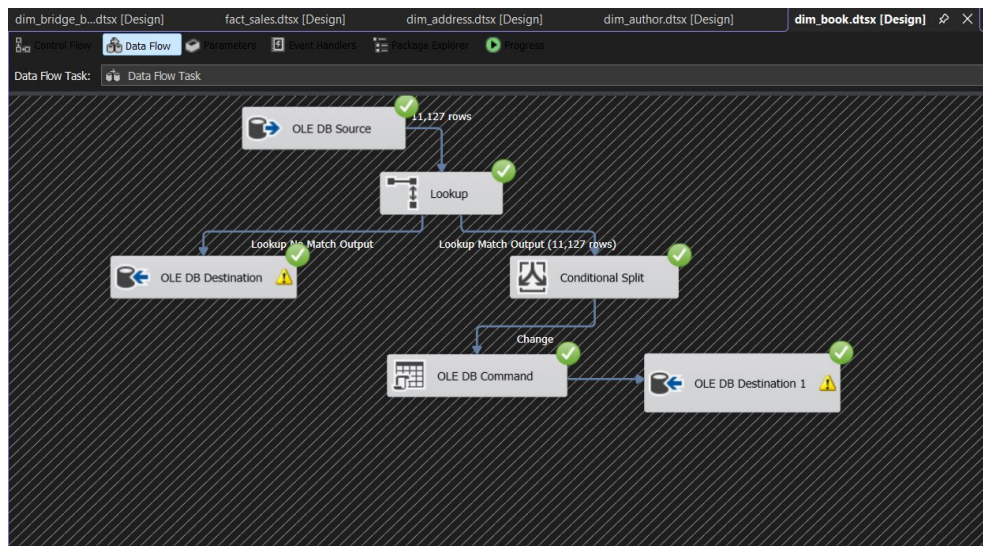


Figure — dim_book.dtsx — SCD2 Data Flow



Figure — `dim_customer.dtsx` — SCD2 Data Flow

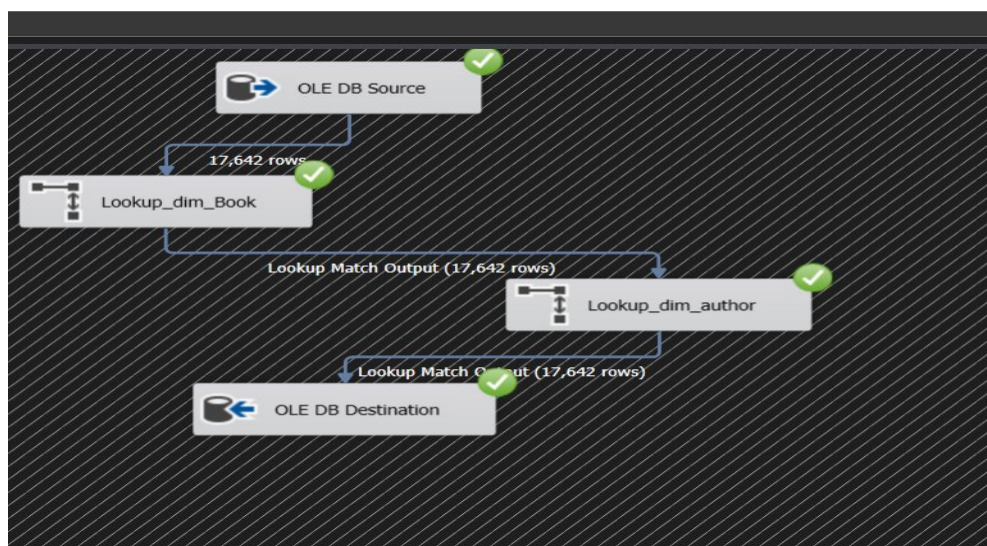


Figure — `dim_bridge_book_author.dtsx`

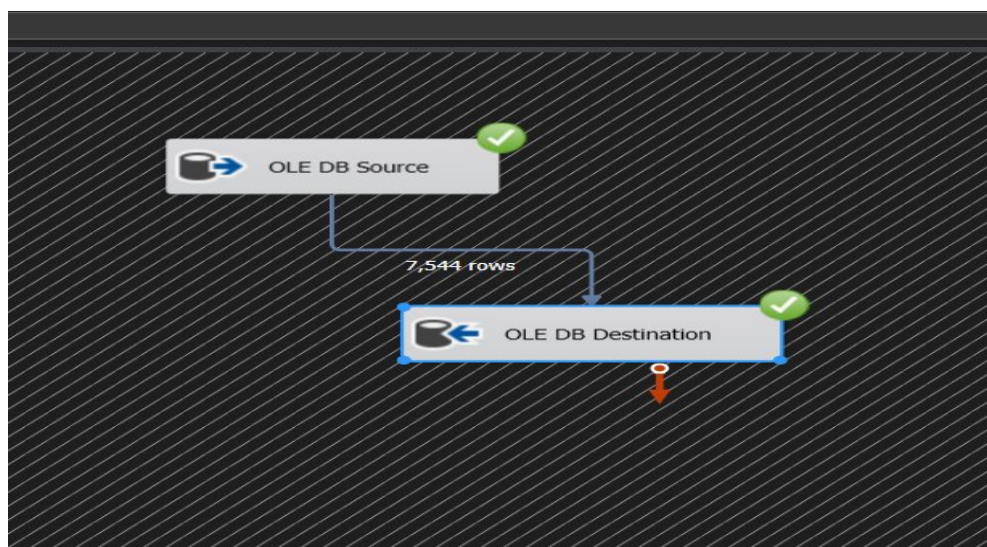


Figure — `dim_status_history.dtsx`

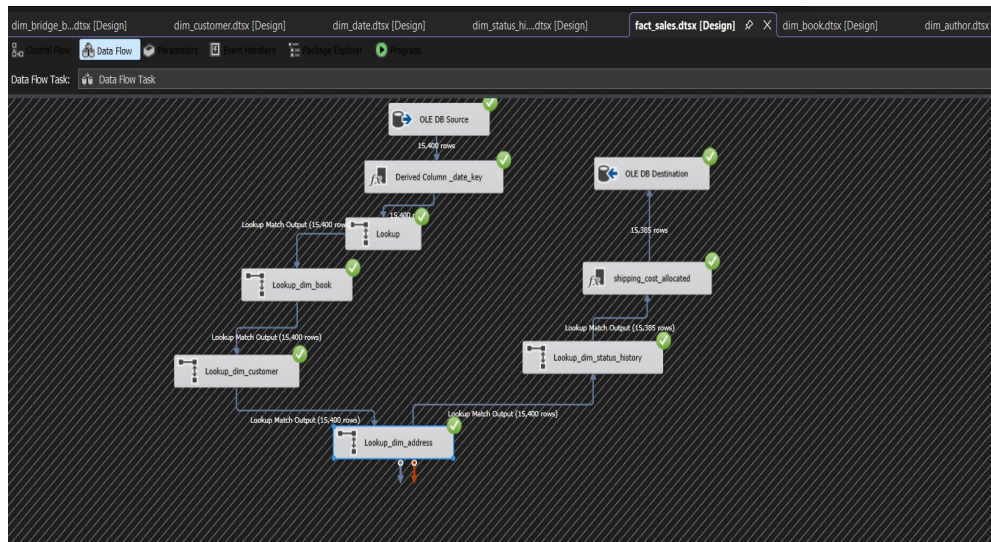


Figure — `fact_sales.dtsx` — 5 Lookups + Derived Column

SCD Type 2 — Data Flow Logic

dim_book and dim_customer use the following pattern to track historical changes:

```
[OLE DB Source: OLTP]
  ↓
  [Lookup: DWH dim]
    ↓ No Match      ↓ Match
  [Insert New]    [Conditional Split]
                   ↓ Changed    ↓ Same
                [OLE DB Command] [Ignore]
                UPDATE is_current=0
                SET valid_to=TODAY
                  ↓
                [OLE DB Destination]
                INSERT new record
                is_current=1, valid_from=TODAY
```

Key SQL in OLE DB Command (Expire old record):

```
UPDATE dwh.dim_customer SET is_current = 0, valid_to = CAST(GETDATE() AS DATE) WHERE customer_sk = ?
— Parameter_0 maps to lkp_customer_sk
```

6. Data Verification

After running all packages, the following record counts confirm successful data load:

Table	Record Count	Notes
dwh.dim_date	7,671	Generated 2015–2035
dwh.dim_author	9,235	All authors loaded
dwh.dim_book	11,127	is_current = 1 for latest
dwh.dim_customer	2,000	is_current = 1 for latest
dwh.dim_address	—	All delivery addresses
dwh.dim_status_history	7,544	Latest status per order
dwh.bridge_book_author	17,642	All book-author links
dwh.fact_sales	15,385	All order lines

Verification Query Sample:

```
SELECT TOP 10
    b.title,
    c.first_name + ' ' + c.last_name AS CustomerName,
    f.price
FROM dwh.fact_sales f
JOIN dwh.dim_book    b ON f.book_sk_fk    = b.book_sk
JOIN dwh.dim_customer c ON f.customer_sk_fk = c.customer_sk;
```

7. SSAS Cube — Bookstore_DWH_View

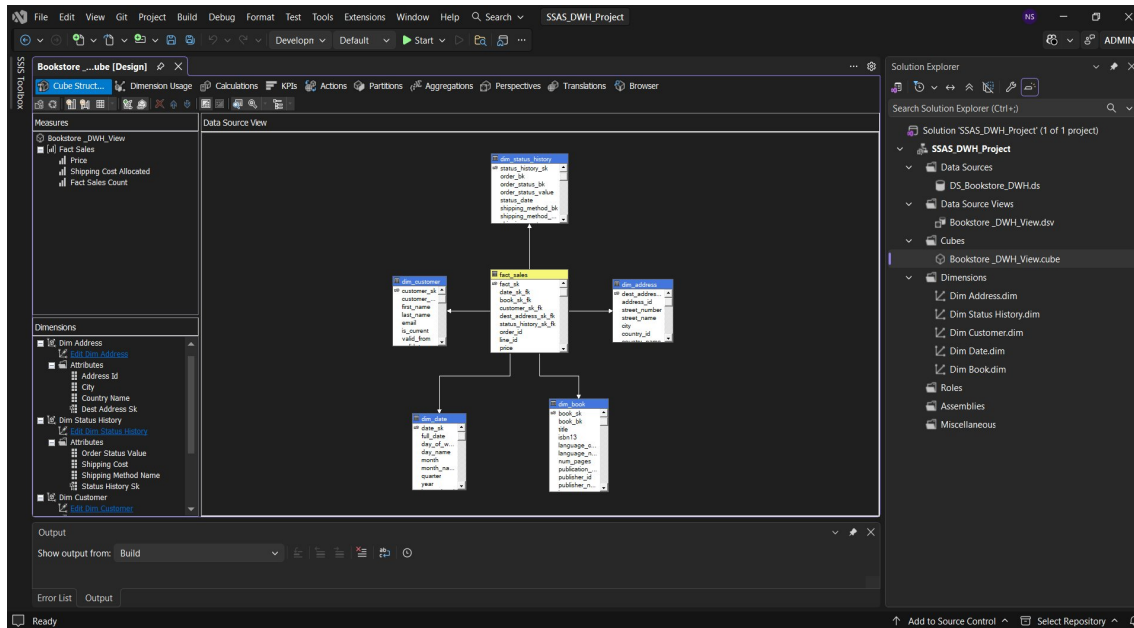


Figure — SSAS Cube Structure showing Fact Sales and all Dimensions

Cube Components:

Component	Name	Details
Data Source	DS_Bookstore_DWH	Connection to Bookstore_DWH SQL Server database
DSV	Bookstore_DWH_View	Includes all 8 DWH tables with relationships
Cube	Bookstore_DWH_View.cube	Built on fact_sales — 15,385 rows processed
Measure Group	Fact Sales	Price, Shipping Cost Allocated, Fact Sales Count
Dim	Dim Date	Hierarchy: Year → Quarter → Month → Date
Dim	Dim Book	Attributes: title, language_name, publisher_name
Dim	Dim Customer	Attributes: first_name, last_name, email
Dim	Dim Address	Hierarchy: Country → City
Dim	Dim Status History	Attributes: order_status_value, shipping_method_name

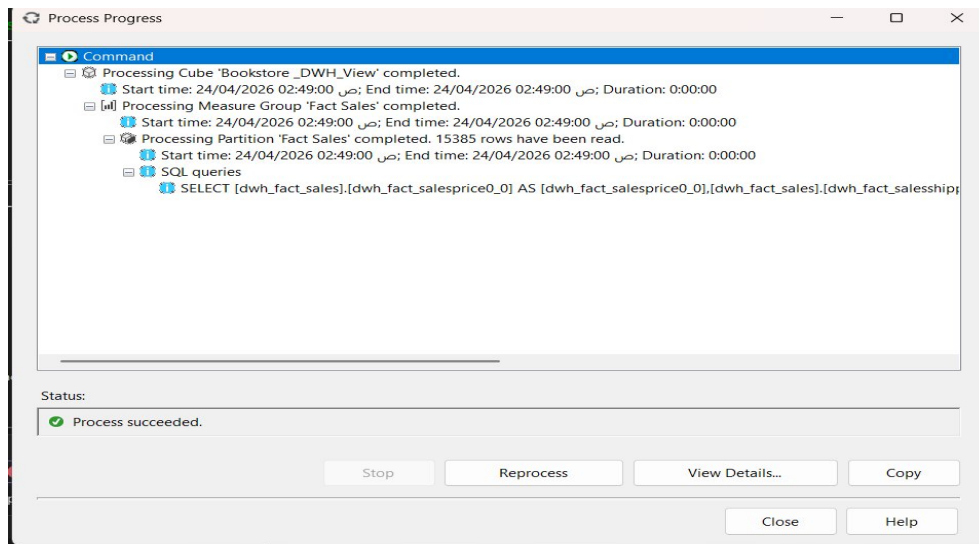


Figure — SSAS Process succeeded — 15,385 rows read from fact_sales

8. Sample Analytical Queries

Total Revenue by Year

```
SELECT d.year, COUNT(DISTINCT f.order_id) AS orders, SUM(f.price) AS revenue
FROM dwh.fact_sales f
JOIN dwh.dim_date d ON f.date_sk_fk = d.date_sk
GROUP BY d.year ORDER BY d.year;
```

Top 10 Best-Selling Books

```
SELECT TOP 10 b.title, b.publisher_name,
             COUNT(f.fact_sk) AS times_sold, SUM(f.price) AS revenue
FROM dwh.fact_sales f
JOIN dwh.dim_book b ON f.book_sk_fk = b.book_sk
WHERE b.is_current = 1
GROUP BY b.title, b.publisher_name
ORDER BY revenue DESC;
```

Top 10 Authors by Revenue

```
SELECT TOP 10 a.author_name,
             COUNT(DISTINCT f.order_id) AS orders, SUM(f.price) AS revenue
FROM dwh.fact_sales f
JOIN dwh.dim_book b          ON f.book_sk_fk  = b.book_sk
JOIN dwh.bridge_book_author ba ON b.book_sk    = ba.book_sk_fk
JOIN dwh.dim_author a        ON ba.author_sk_fk = a.author_sk
WHERE b.is_current = 1
GROUP BY a.author_name ORDER BY revenue DESC;
```

Sales by Shipping Method

```
SELECT s.shipping_method_name, s.order_status_value,
             COUNT(DISTINCT f.order_id) AS orders,
             SUM(f.shipping_cost_allocated) AS shipping_revenue
FROM dwh.fact_sales f
JOIN dwh.dim_status_history s ON f.status_history_sk_fk = s.status_history_sk
GROUP BY s.shipping_method_name, s.order_status_value
ORDER BY orders DESC;
```

9. Architecture Summary

OLTP: gravity_books (SQL Server)

■■■ 13 normalized tables

■■■ Transactional data – books, orders, customers

↓ SSIS ETL (8 packages)

DWH: Bookstore_DWH (SQL Server)

■■■ Schema: dwh

■■■ 1 Fact Table : fact_sales (15,385 rows)

■■■ 5 Dimensions : date, book, customer, address, status_history

■■■ 1 Bridge Table : bridge_book_author

■■■ 1 Outtrigger Dim : dim_author

↓ SSAS

Cube: Bookstore_DWH_View.cube

■■■ Measures : Price, Shipping Cost Allocated

■■■ Dimensions: Date (with hierarchy), Book, Customer, Address, Status

■■■ Ready for: Excel Pivot Tables / Power BI

Project Status: All 8 SSIS packages executed successfully. SSAS Cube deployed and processed with 15,385 rows. DWH is ready for reporting and analysis.